# The Kepler Initiative — Third-Year Progress Report

The Kepler Initiative is a tidal-monitoring programme run jointly by Meridian Labs and the Halden Institute. This report covers its third year of operation, ending in 2025.

## Summary

Kepler now operates forty-one monitoring buoys along the Atlantic shelf, up from twenty-eight at the end of the previous year. All units are Wavelet platforms built in Tromsø. The programme has published continuous readings since its second year and its dataset is used by fourteen external research groups.

## Governance

The programme is co-chaired by A. Okonkwo of Meridian Labs and by Dr. Ingrid Halvorsen, who directs the Halden Institute. The two organisations share operating costs equally. Day-to-day coordination runs out of Meridian's Lisbon office.

Priya Raghunathan serves as technical lead. Her team handled the migration of the buoy fleet to the second-generation Wavelet unit during the reporting period.

## Findings

The most significant result this year concerns the Cascais Anomaly — a persistent temperature gradient observed off the coast near Lisbon, first identified by Okonkwo in the programme's second year. Extended observation has confirmed the gradient is seasonal rather than permanent, which contradicts the initial hypothesis published at the start of the programme.

A second finding concerns instrument drift. Wavelet units deployed for more than eighteen months show measurable salinity drift, and the team now recommends recalibration on an annual cycle.

## Funding and outlook

Kepler is funded through the Nordic Marine Council, with additional support from the Costa Foundation. Funding is committed through the programme's fifth year. The partners intend to expand the network to sixty buoys, contingent on the Council renewing its grant.